Lab 7 PreLab

March 26, 2025 4:37 AM

P1. Consider the CE amplifier of **Figure 1**, driving a load R_L. Assume that $R_1 = 12\ k\Omega$, $R_2 = 15\ k\Omega$, $R_C = 2.7\ k\Omega$, $R_E = 6.8\ k\Omega$, $R_{E2} = 220\ \Omega$, $V_{CC} = 15\ V$, $\beta = 150$, $V_{BE,on} = 0.7\ V$, and $V_{CE,sat} = 0.3\ V$. Also assume the capacitors to be short at the operating frequencies of interest. Then, manually calculate the DC (quiescent) currents and voltages of the amplifier, as well as the no-load voltage gain A_{vo}, voltage gain $A_v = v_o/v_i$ (with $R_L = 180\ \Omega$), voltage gain $A_{vs} = v_o/v_s$ (with $R_s = 50\ \Omega$ and $R_L = 180\ \Omega$), and output resistance R_o. Complete **Table P1**. Show all work.

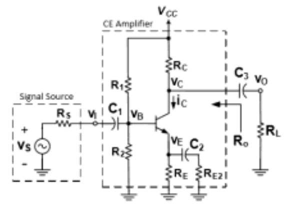


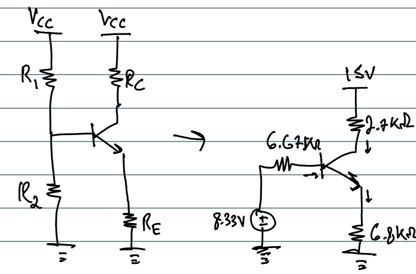
Figure 1. A Common-Emitter (CE) amplifier driving a load directly.

Table P1. Quiescent and AC parameters of the CE amplifier of **Figure 1**.

$V_C[V]$	$V_E[V]$	$I_C[mA]$	$A_{vo}[V/V]$	$A_v[V/V]$ $R_L = 180\ \Omega$	$A_{vs}[V/V]$ $R_L = 180\ \Omega$ $R_s = 50\ \Omega$	$R_o[k\Omega]$
12	7.54	1.1	-57	-3.56	3.56	2.7

$R_1 = 12k\Omega$
$R_2 = 15k\Omega$
$R_C = 2.7k\Omega$
$R_E = 6.8k\Omega$
$R_{E2} = 220\Omega$
$V_{cc} = 15V$
$\beta = 150$
$V_{BE,on} = 0.7V$
$V_{CE,sat} = 0.3V$
$R_L = 180\Omega$

Calculate DC values:



$R_1 \| R_2 = 6.67k\Omega$

$V_{BB} = \frac{R_2 \cdot V_{cc}}{R_1 + R_2} = 8.33V$

$-8.33 + 6.67k i_b + 0.7 + 6.8k i_e = 0$

$6.7k i_b + 6.8k(\beta+1) i_b = 8.33 - 0.7$

✓ $i_b = \frac{7.63V}{(6.7k\Omega + 6.8k(151))}$

$= 7.38uA$

✓ $i_e = i_b(\beta+1)$

$= 1.11mA$

✓ $i_c = \beta i_b = 1.10mA$

$R_o = R_C$

$= 2.7k\Omega$

$-V_{cc} + 2.7k i_c + V_c = 0$

✓ $V_C = V_{cc} - 2.7k i_c$

$= 12V$

✓ $V_e = I_E R_E$

$= 7.54V$

$-V_e - 0.7 + V_b = 0$

✓ $V_b = 8.24V$

$V_{ce} = 4.46V$

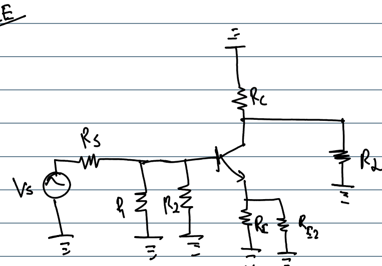
✓ $V_{bc} = V_b - V_c = -3.76 < 0.7$

✓ $V_{be} = 0.7V$

∴ Active mode verified

AC mode:

CE



$A_{v_o} = \frac{-g_m(R_C)}{1 + g_m R_e} = \frac{-113.4}{1.99}$

$= -57.10\ V/V$

$A_v = \frac{-g_m(R_C \| R_L)}{1 + g_m R_e}$

$= \frac{7.09}{1.99} = -3.56$

$g_m = \frac{I_c}{V_T}$

$= 42ms$

$r_\pi = \frac{\beta}{g_m} = 3.55k\Omega$

$r_e = \frac{r_\pi}{\beta+1} = 23.48\Omega$

P2. Now consider the two-stage amplifier of **Figure 2**, in which the **CE amplifier** of **Figure 1** is not directly connected to the load, but it **drives the load through a Common-Collector (CC) amplifier**. Thus, **the CC amplifier acts as a buffer** between the CE amplifier and the load. That is, it makes the load appear much larger to the CE amplifier, to not bring down the output voltage of the CE amplifier, while its own voltage gain is close to (albeit smaller than) unity. Then, repeat the analysis of **Step P1** for the amplifier chain of **Figure 2**. Assume the same parameters as those you assumed in **Step P1**, with the addition that $R_{E3} = 1.2\ k\Omega$. Complete **Table P2**. Show all the work.

$R_{E3} = 1.2k\Omega$
$R_L = 180\Omega$

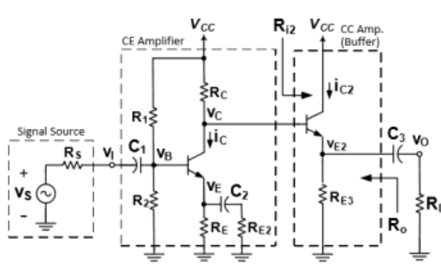
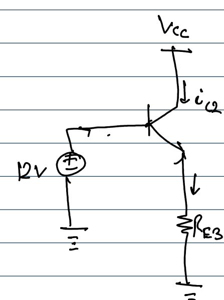


Figure 2. Common-Emitter (CE) amplifier of Figure 1 driving a buffered load.



$-12V + 0.7V + R_{E3} I_{E2} = 0$

$-12V + 0.7V + R_{E3} I_B(\beta+1) = 0$

$I_B = \frac{12V - 0.7V}{R_{E3}(\beta+1)}$

$= 62.36uA$

$I_E = (\beta+1) I_B = 9.42mA$

$I_{C2} = \beta I_B = 9.35mA$

Table P2. Quiescent and AC parameters of the two-stage amplifier of **Figure 2**.

$V_C[V]$	$V_E[V]$	$I_C[mA]$	$V_{E2}[V]$	$I_{C2}[V]$	$A_{vo}[V/V]$	$A_v[V/V]$ $R_L = 180\ \Omega$	$A_{vs}[V/V]$ $R_L = 180\ \Omega$ $R_s = 50\ \Omega$	$R_o[k\Omega]$
12	7.54	1.1	11.3	0.009	0.99	0.996	0.996	2.75

$V_{C2} = V_{cc} = 12V$

$V_{E2} = R_{E3} \times I_E$

$= 11.30V$

$A_v = \frac{g_m(R_e \| R_L)}{1 + g_m(R_e \| R_L)}$

$= \frac{156.19}{1 + 156.?} = 0.996V/V$

$r_\pi = \frac{\beta}{g_m} = 417.82\ \Omega$

$g_{m_2} = \frac{I_{c2}}{V_T} = 359ms$

$r_e = r_\pi$ ...

$r_\pi = \frac{D}{g_m} = 417.82\ \Omega$

$r_e = \frac{r_\pi}{B+1} = 2.76\Omega$

$g'_{m_2} = \frac{I_{C2}}{V_T} = 359ms$

$R_e = R_{e3} + r_e$

$= 2.76\ \Omega + 1.2k\Omega$

$= 1.202k\Omega$

$= \frac{156.19}{1+156.?} = 0.996 V/V$

$A_{vo} = \frac{g_m R_e}{1+g_m R_e} = \frac{430.8}{1+430.8} = 0.99 V/V$

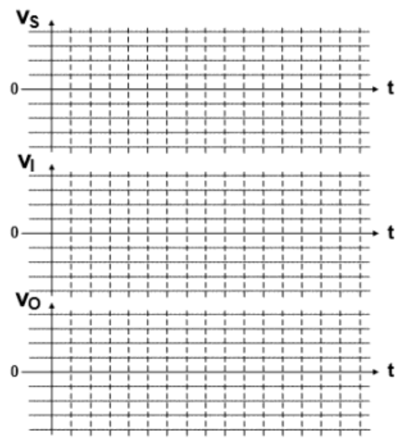
$R_o = R_e \| r_e$

$= 2.75\ \Omega$

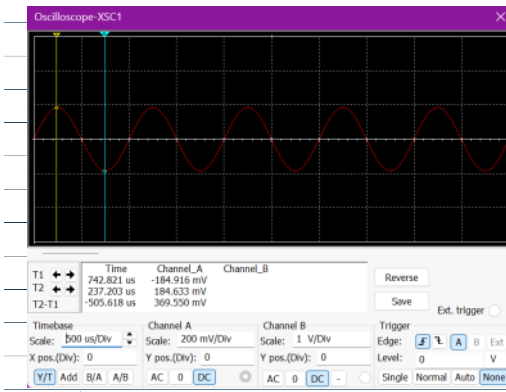
$V_o = 2V$ pk-pk

$V_{S,max} = \frac{V_{o,max}}{|A_V|} = \frac{2V}{0.996} = 2.008$

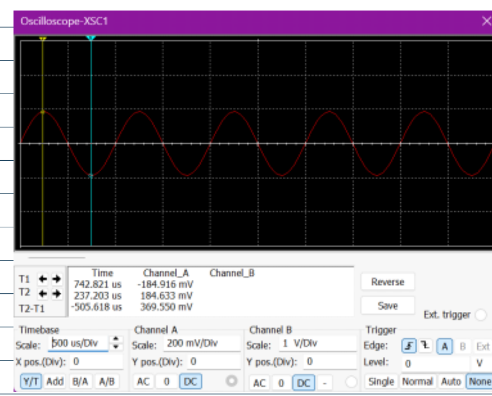
P3. Simulate the CE-CC two-stage amplifier of **Figure 2**, assuming the 2N3904 as the transistors, $V_{CC} = 15\ V$, $R_S = 50\ \Omega$, $C_1 = 10\ \mu F$, and $C_2 = C_3 = 100\ \mu F$. Also, assume v_S to be a **1-kHz symmetrical sinusoidal voltage.** However, choose its magnitude in such a way that, with no load (for the purpose of simulation, use $R_L = 100\ k\Omega$ as an infinite load resistance), v_O has a peak-to-peak swing of 2.0 volts. Then, present the waveforms of v_S, v_I, and v_O, for three cycles, for $R_L = 100\ k\Omega$, as **Graph P3(a)**. Next, repeat the simulations but with $R_L = 180\ \Omega$, and present the waveforms as **Graph P3(b)**. In both simulations, make sure that v_I and v_O are in-phase (in terms of their zero- crossings). Otherwise, check your simulation model and parameters. Also, verify that your manual calculations of the gains (reported in **Table P2**) agree well with the gains indicated by the simulated waveforms. Otherwise, check your calculations and/or your simulation model.



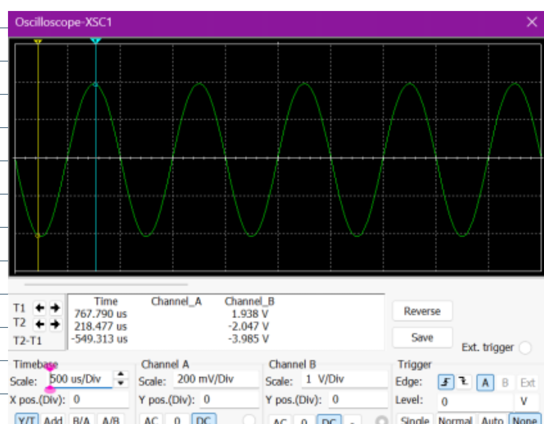
Graph P3(a). Source, input, and output voltage waveforms of the amplifier of Figure 2, with $R_S = 50\ \Omega$ and $R_L = 100\ k\Omega$.



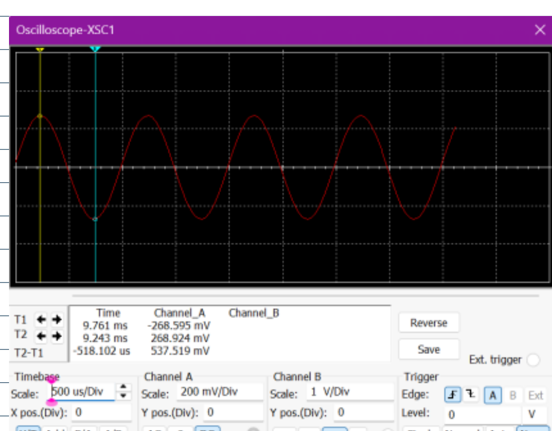
P3(a): Vs vs T

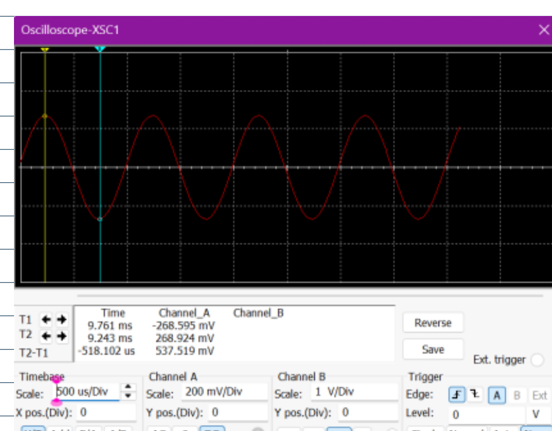


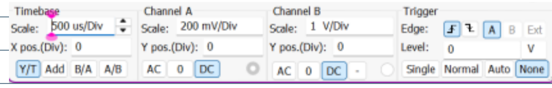
P3(a): V_I vs T



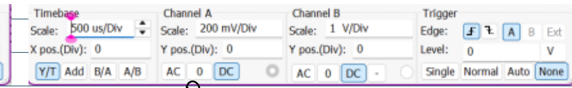
P3(a): V vs T



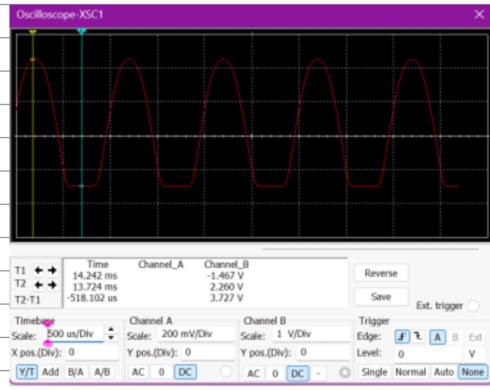




P3(b): Vs vs T



P3(b): V_I vs T



P3(b): V_o vs T